Memory Is Not Enough: Separating Drift from Background Conflict in Continual Document Understanding

Anonymous CVPR submission

Paper ID *****

Abstract

Replaying a few hundred stored documents nearly matches joint training when a continual key-information extractor meets new domains, yet collapses when its label space grows. We ask why, as an investigation rather than a method. On a class-incremental stream of receipts we test a ladder of hypotheses, each with its prediction stated in advance, and measure where the errors go. Forgetting is not simply harder: with a whole-document memory and a shared softmax, the old entities it still predicts are mostly right, but most are missed, and the newest sessions are barely learned. Stale stored labels explain part of it, but no memory we tried removes the failure: giving the older stored documents their true labels at each boundary still leaves the model below joint training (one seed). What removes it is to stop sharing the background class, as existing per-session readouts do: with the same memory and no true labels, one background logit per session reaches 89.1 *full-label F1 against* 92.0 *for joint training. The principle, familiar from training on partially annotated data, is that an "O" label is evidence only against the classes its annotator could see. Where label growth supplies the routing key, on a single-schema stream, per-session low-rank trunk updates can then replace the memory; across schemas they cannot.*

1. Introduction

Key-information extraction (KIE) labels each OCR token of a visually-rich document (form, invoice, receipt), using its words, its 2-D layout and the page image; multimodal encoders such as LayoutLMv3 [17], LiLT [49] and BROS [15] define the state of the art. Deployed extractors meet new document types and growing schemas, while retraining on all history is blocked by cost and by privacy constraints: a continual-learning (CL) problem, whose central obstacle is catastrophic forgetting [32].

Replay is empirically the strongest remedy in the CL toolbox [31, 50], and on documents it looks sufficient. On LayoutLMv3, a stream of forms and receipts under one fixed label space is learned by plain replay (ER) with 200 stored documents to 87.9 AA, against 88.7 for joint training (Tab. 3). Let the label space grow instead, by adding CORD's 30 receipt fields six at a time, and the same method (200 documents) reaches 24.3 AA, against 91.3; both ER cells are legacy-protocol runs (†) that early-stop on the test split, which favours ER (Sec. 3). Distillation, prompts, low-rank adapters and the continual-NER methods built for the shift of the "O" label do no better (Tab. 4). A class-incremental stream is expected to be harder; what is not obvious is why a memory that holds the old documents does not help. This paper is about that question: *why does replay, which solves domain shift, fail when the label space grows?*

We treat it as an investigation (Tab. 1): we write down the explanations a reader would offer, state what each predicts before its test, test each with an intervention that changes one thing on the same encoder, stream and recipe, and measure where the final model's errors go (Tab. 2). Forgetting is not just harder (H0): a whole-document memory that retains under domain shift misses most old entities and suppresses the session being trained, which drift does not do. Stale labels in the memory (H1) are part of the cause: relabelling the older stored documents at each boundary recovers much of the gap but stays below joint training and below the same memory with the readout introduced here (one seed). Label-side corrections of what current documents say about old entities (H2) recovered part of the gap; a MiB-style unbiased loss on the same memory is the decisive test of that rung. Removing the shared background (H3) closes most of the gap with no true labels, given a memory, and then per-session isolation can replace the memory (H4), but only where label growth supplies a routing key.

None of the ingredients is new. The background shift is known from class-incremental segmentation [4, 6, 9], from incremental detection, where both its stored side [27] and its two-sided form [34] have been named, and from continual NER [29, 35, 55]; separated softmaxes [1], per-task heads with a "none of mine" output [19] and per-session softmaxes over a constant Other logit [26] already undo the shared nor-

malisation; forgetting in the output-facing layers is known from image classifiers [39]. What has not been done is to take apart, on token-labelled documents, why replay fails: which part a memory repairs, which part survives a memory with true labels, and what the errors of each look like.

Contributions.

1. **The investigation.** A decomposition of why replay fails under label growth that prior work did not run on token-labelled documents: a memory × readout factorial, the stored-label side removed by a relabelled memory with the gap that remains, and direct measurements that tell the causes apart (drift sends old entities to another session's class through over-claiming blocks; the conflict leaves them missed and suppresses the session being trained; Sec. 4). The relabelled-memory control rests on one seed so far.
2. **An empirical finding, not a new mechanism.** A per-session background readout, as in ICE-O and MORE, combined with whole-document replay and a supervision rule in the spirit of training on disjoint tag sets, closes most of the gap to joint training in continual KIE on every seed, where the continual-NER and replay methods in our comparison do not. Our instantiation (SBR) differs from ICE-O in a learned background per session and in old blocks kept trainable from the memory (Sec. 5).
3. **Boundary conditions.** With the conflict removed, per-session low-rank trunk deltas (SBR-Recall) replace the memory on a single-schema stream, within 2.1 points of joint training on every seed under both metrics, and fail without a memory across annotation schemas (one seed), where the task-identity view says they must (Sec. 5.3).

2. Related Work

Background shift. When each step annotates only its own classes, the background label changes meaning. In class-incremental segmentation MiB [4] matches a background label with the probability of background or any old class in the loss, and the old model's background with background or any new class in the distillation; PLOP [9] and SSUL [6] pseudo-label the background with the old model, and SSUL also freezes the backbone and the past classifiers, keeps a tiny memory and uses per-class sigmoids. Incremental detection has named both sides: replayed images carry current objects in their background (foreground shift [27]), and past and future objects are both labelled background, so the objectives of different stages conflict [34]. In continual NER the shift of "O" is the unlabelled-entity problem [29], answered by distillation on "O" tokens [35, 59], pseudo-labels with feature distillation [55], contrastive learning with relabelling [29], entity-triplet generation [53], or one binary output per type [57]. We do not claim the conflict; we separate its stored and current sides with a relabelled memory and measure what each leaves.

Separated readouts and task identity. A softmax over old and new classes together biases them, and separating it removes the bias [1]. Giving each task its own "none of mine" output turns class-incremental learning into within-task prediction plus task identification [20]: MORE [19] trains a head per task with an out-of-distribution output on saved samples and lets the most confident head answer, and in class-incremental extraction ICE-O [26] trains, on a frozen encoder and with no memory, one softmax per session over a constant Other logit and that session's types, so that a mention is Other only if every session scores below that logit. SBR is this pattern on documents with a learned background per session and old blocks kept trainable from a memory; SBR-Recall adds per-session trunk deltas. Adapter methods instead merge or combine per-task modules so that no task identity is needed [25, 51, 60]. Outside CL, taggers trained on datasets with disjoint tag sets either marginalise over the tags a dataset does not annotate [11] or share one tagging layer through a tag hierarchy [2]; the first is the loss MiB uses, and we make no claim that separation beats it in general.

Latent and compressed replay. REMIND [13], Latent Replay [36] and generative replay [48] replay stored or generated mid-network representations through the plastic upper layers. Exemplar-free methods keep class statistics: PASS and FeTrIL generate old-class features from prototypes [37, 61], FeCAM classifies by a per-class Mahalanobis distance on a frozen backbone [10], SLCA realigns the classifier with class Gaussians [56], and in segmentation replaying class prototypes rivals image replay [7]. A document carries one label per token, bound to its position and its page; our memory keeps that with one truncated SVD per document, a per-sequence factorisation used to compress key caches [46], here used as a replay memory.

Where forgetting lives; continual document IE. Ramasesh et al. [39] localise forgetting in image classifiers to the layers nearest the output; Kalb and Beyerer [18] find that in domain-incremental segmentation freezing early layers moves the representation shift to later layers rather than removing it. Continual relation extraction [8, 12, 58], like continual NER, is sentence-level and text-only. Document-level continual learning is recent and small: GDI-Bench [23] limits forgetting in multimodal-LLM fine-tuning by freezing most parameters; continual table detection [33] uses plain ER; ProtoNER [22] adds key-information classes to a LayoutLM-family model from a few documents with prototypical networks; AMD-Proj [44] projects gradients against a memory of task subspaces on FUNSD, SROIE, CORD and

BuDDIE; and Sassioui et al. [43] benchmark forgetting of layout-aware models on the same datasets by stream regime (domain transfer, layout drift, schema evolution), finding that no method they test preserves old knowledge under label-space shift. They decompose forgetting by regime of the stream; we decompose it by cause, within label growth. The broader CL families [3, 21, 24, 28, 30, 40, 42, 45, 52] are surveyed by Wang et al. [50].

3. Setup

Task and streams. A document is a triple of OCR tokens, normalised boxes and the page image; the model emits BIO tags per token and is scored by entity-level F1 (`seqeval`). We use FUNSD (149/50 train/test forms), CORD (800/100/100 receipts, 30 classes) and SROIE (626/347 receipts). **DIL**: FUNSD→SROIE→CORD under one fixed four-entity schema; only the input distribution shifts. **CIL-CORD**: CORD's 30 classes arrive six at a time over five sessions; the head grows, and tokens of every class outside the session, earlier or later, are labelled "O". **Mixed**: six sessions over three datasets and annotation schemas (two class-incremental FUNSD sessions, a shift to SROIE, two class-incremental sessions on CORD's super-classes, a FUNSD revisit with all its labels). App. A gives the sessions.

Backbones, memory and comparison set. LayoutLMv3-base [17] is the primary encoder; LiLT [49], BROS [15] and text-only BERT are secondary. Heads are linear. Every memory-based run of the investigation uses one whole-document latent memory (CoLaR, Sec. 5.2): after the first session the embeddings and layers $1..4$ are frozen, and 50 documents per session (per task on DIL) are stored as a rank-128 truncated SVD of their layer-4 activations with their boxes and labels and replayed through the upper layers, four per step. The comparison set is fixed in advance: naive fine-tuning and joint training as references, and one published method per family: ER [41] and DER++ [3], LwF [24], CODA-Prompt [45], CL-LoRA [14], LCA [47], and the continual-NER methods CPFD [55] and IS3 [38]. Replay baselines keep 200 raw documents (DER++ $\alpha{=}\beta{=}0.5$); in clean-protocol cells ER instead keeps as many as our memory. LwF uses $T{=}2$. Our CL-LoRA port trains a single rank-8 adapter on the attention projections of the frozen encoder, shared by all sessions, unlike the published design, which pairs task-shared and task-specific adapters; its numbers describe this port.

Training protocol. AdamW, learning rate $5{\times}10^{-5}$, weight decay 0.01, gradient clipping 1.0, batch size 2, seeds 42, 7 and 123. Under the *clean protocol* every session trains in full precision for at most 10 epochs (joint training: 100), with early stopping (patience 2) on a 10% split held out from that session's *training* data and scored with the same session restriction as the test, so a prediction of another session's class is neither rewarded nor charged; the test split is never used for model selection. Cells marked † come from earlier runs (the *legacy protocol*), which early-stop on the session's test split (except joint training on Mixed) and vary in epoch cap (10 to 100) and precision. A cell switches to the clean protocol only when all three clean seeds exist and never combines the two; App. G lists the settings of every † cell.

Scoring. The headline number is the *full-label F1*: the final model on the complete CORD test set with all 30 classes, the usual evaluation of continual NER [29]. Only clean runs record it; a comparison with a cell that has only AA († cells, DIL, Mixed) is made in AA on both sides. Retention needs per-session test sets, whose labels show only that session's classes; from the retention matrix $R[i, j]$ (F1 on session j after session i) we report AA $= \frac{1}{T}\sum_j R[T, j]$. Scoring raw predictions against such labels is a pitfall: a model that correctly labels another session's entity is charged a false positive in every session that does not own it, so a jointly trained model reads 33 (App. B). We score each session *session-restricted*: a predicted tag outside session j's label set becomes "O" before span matching, so an error is charged at most once. The two metrics weight differently: AA averages the sessions, while session 1 holds 57% of the 1,306 test spans. Pooling a final model's per-session span counts reproduces its full-label F1 within 0.4 points, except for the blocked readout without memory, whose false positives on other sessions' receipts the splits omit; on every seed both metrics order the groups of clean runs the same way (App. B).

How the settings were chosen. Tuning and evaluating on the same scenario overstates CL methods [5], so we state how each setting was fixed. The comparison set was fixed before the class-incremental runs, one method per family. The memory settings ($k{=}4$, $r{=}128$, 50 documents) were chosen on DIL with LayoutLMv3 (seed 42; App. F) and carried unchanged to every class-incremental stream. SBR's claim threshold is 0.5 and was not tuned (tuning it on the test set moves AA by at most 0.6; Sec. 6). The one setting chosen on CIL-CORD is SBR-Recall's delta capacity: rank 8 on the attention projections left two sessions under-learned in legacy runs that early-stopped on the test split, and we adopted rank 64 with the feed-forward layers before the clean re-runs (App. D). We also built and dropped five buffer-free families, memory summaries, read-side memory additions, rebalancing, lexical routing, budget allocation, public-proxy replay and an SBR variant without its cross-schema rule; App. F lists each with its numbers and failure reason.

4. The investigation

Every rung below runs on CIL-CORD with LayoutLMv3 under the clean protocol unless stated, and changes one thing against the run it is compared with. Table 1 lists the rungs and Tab. 2 where the final models' old-session errors go. Each rung states its prediction before its result, and a verdict word is one the numbers justify on every seed.

4.1. H0: is it just harder forgetting?

Prediction. If label growth only made drift harder, the memory that retains under domain shift would still retain here, a raw memory would do better, and the losses would look like naive forgetting, where old sessions are lost and the newest is learned well.

Test and result: refuted. The CoLaR memory comes within a point of joint training on DIL on average (AA 87.8 against 88.7, Tab. 4). On CIL-CORD with a shared softmax it reaches 30.7 ± 1.5 full-label F1 (33.5 ± 1.0 AA) against 92.0 ± 0.4 (91.3 ± 0.8), and raw documents do no better (ER, 24.3 AA†, below every CoLaR seed). The errors do not look like naive forgetting. The full-label F1 and the exact span counts imply a precision of 71.8 to 79.2% over all sessions, but the model finds only 18.7 to 22.6% of the old sessions' spans (Tab. 2); a span-level bound puts at least 89% of its missed spans entirely at "O" (an indicator: its assumption fails on one blocked run, App. B). And the newest session ends at 2.0 to 16.0 F1, whereas naive training, which forgets every old session, learns every session to at least 78.6. With a memory the model mostly misses old entities rather than confusing them, and it misses those of the session it is training too.

What it rules out, and what it leaves. Memory fidelity is not the bottleneck, since raw documents do no better. Drift is still real: naive training keeps nothing of the old sessions, and on a fixed label space forgetting sits in the output-facing computation on all four encoders, as known for image classifiers [39]; freezing the head and late layers moves the drift to the early layers rather than removing it (one seed), the mirror image of what Kalb and Beyerer [18] report for segmentation. On token-level documents only old-task supervision of the readout repaired it in our experiments, whereas on images class statistics can [56]; Tab. 3 and App. F hold the evidence.

4.2. H1: are the stored labels stale?

Prediction. A document stored at session t carries "O" on every entity of another session, earlier or later, so the memory itself calls background what the model must keep or learn. In detection the same staleness of replayed examples was named foreground shift and blamed for replay's failure [27]. If these stale labels are the cause, giving the stored documents their true labels should close the gap with the shared softmax unchanged.

Test. At the end of each session we give every document stored *before* it the true labels of all classes seen so far. The documents of the session just finished keep their own labels until the next boundary, and no refresh can cover the session trained next. The refresh rewrites $105/1019/1394$ stale tokens before sessions 3 to 5 (and 1239 after the last session, which changes no training; seed 42). It reads labels no continual learner has and is a diagnostic, not a method.

Result: partly. Against the same memory without the refresh it lifts full-label F1 from 32.4 to 79.1 and AA from 33.6 to 65.8 (seed 42, one seed so far): 79% and 56% of the seed-matched gap to joint training (91.7, 91.1), and still below CoLaR-SBR (88.8, 87.8) under both metrics. The metrics disagree because of what the refresh cannot reach. While session t trains, its own classes are "O" in every stored document, and sessions 3 to 5 are learned only to 2 to 15 F1 on their own diagonal; sessions 3 and 4 rise to 74 to 84 one session later, once their own documents are stored and the older ones carry their labels, and the last session, which has no later boundary, ends at 14.7. AA weights it as a fifth of the score; it holds 8% of the test spans.

What it rules out, and what it leaves. Stale stored labels are a large part of the failure, and they act on the session being trained as much as on old ones: a session suppressed while current and recovered one boundary later is the stale-label mechanism seen directly, and the limit of this control. Whether a memory that is never stale closes the gap within a shared softmax is what relabelling every stored document *before* each session tests; that control leaves the current documents as the only stale source.

4.3. H2: do current documents mislabel old entities?

Prediction. A current document labels every entity of an earlier session "O", so even a memory with true labels is opposed in every batch by documents that call the same strings background. This is the second side of the conflict named in detection [34]. If this pull is the cause, a loss that lets a current "O" mean "O or an earlier class", as MiB does for segmentation [4] and marginal-likelihood taggers do offline [11], should close the gap within a shared softmax; prior work gives no reason to expect a separated readout to do better than it.

Test. The decisive control keeps the memory and the shared softmax and uses MiB's unbiased cross-entropy in

Table 1. **The hypothesis ladder** (CIL-CORD, LayoutLMv3 unless stated; mean ± s.d. over three seeds, per-seed values when fewer; $^{n=k}$: k clean seeds so far; †legacy protocol; "–": no full-label run). Verdict: what the row says about its rung's prediction, on every seed. ^{r}At each boundary the documents stored before the session just finished receive their true labels (Sec. 4.2).

Rung	Hypothesis	Prediction if true	Intervention (CIL-CORD unless stated)	Full-label F1	AA	Verdict
			Naive fine-tuning (reference)	13.0±0.2	18.3±0.5	
H0	Only drift: forgetting is just harder	a memory that retains under domain shift retains here; raw beats compressed	whole-document memory (CoLaR), shared softmax	30.7±1.5	33.5±1.0	refutes
			raw-document memory (ER), shared softmax	–	24.3±2.4†	refutes
H1	Stale labels in the memory	true labels in the stored documents close the gap	older stored documents relabelled at each boundary^{r}	79.1$^{n=1}$	65.8$^{n=1}$	partly
H2	Current documents mislabel old entities	a loss that lets "O" mean "O or an old class" closes the gap	replay and current losses sliced to their classes	–	60.5±10.1†	partly
H3	The shared background normalisation	one background per session closes the gap without true labels, given a memory	CoLaR memory, session-blocked readout (CoLaR-SBR)	89.1±2.0	88.4±2.6	supports
			session-blocked readout, no memory	11.4±2.3	21.6±2.2	supports
H4	Isolation replaces memory where a routing key exists	per-session deltas need no memory on one schema, and fail across schemas	per-session deltas, blocked readout, no memory (SBR-Recall)	92.6±1.0	91.9±0.6	supports
			the same on Mixed (AA), no memory	–	44.7†	supports
			the same on Mixed (AA), 50-document memory	–	85.9†	supports
			Joint training (reference)	92.0±0.4	91.3±0.8	

Table 2. **Where the old-session errors go** (final model, sessions 1 to 4; range over n clean seeds). R, P: recall and precision of old-session spans ($\leq$: an upper bound). Old entity tokens: share claimed by their own session's block (owner); wrong ones predicted "O", given another class of their session, or of another session. False: tokens of a session's split that are not its entities, claimed by its block. Blank: does not apply or not recorded.

		Old spans		Old entity tokens				
Run	n	R	P	owner	→O	same	other	false
CoLaR, shared softmax	3	18.7–22.6	≤72.8–80.8					
Older documents relabelled^{r}	1	77.2	≤88.2					
Blocked readout, no memory	3	0.4–13.2	3.4–10.7	10.6–17.6	0.0	6.5–16.2	83.8–93.5	17.0–24.3
CoLaR-SBR	3	88.4–92.6	84.8–90.4	95.4–98.3	4.9–6.2	56.6–71.6	22.3–38.4	0.6–1.7
SBR-Recall	3	92.7–94.0	90.4–93.2	97.0–98.3	4.4–28.4	51.1–65.2	20.0–30.4	0.4–0.8

Table 3. **Drift and its remedy on four encoders** (DIL, AA, mean ± s.d., three seeds; all cells from our first grid, legacy protocol, App. G). On average, replay comes within one point of joint training or exceeds it, and EWC and LwF stay within ten points of naive training. "–": did not fit the 6 GB GPU.

Backbone	Naive	EWC	LwF	ER	DER++	Joint
LayoutLMv3	41.3±1.5	41.2±1.1	43.7±1.1	87.9±1.4	88.0±0.9	88.7±1.1
LiLT	40.1±0.5	48.4±5.9	–	85.7±0.5	84.4±0.3	83.8±1.2
BROS	40.3±1.8	45.0±4.9	35.1±3.1	87.8±0.4	87.5±1.1	87.5±1.3
BERT	39.4±0.3	46.6±3.0	40.5±0.1	77.1±0.9	77.4±0.6	77.2±0.4

both directions: a current "O" token matches "O" or any earlier-session class, a replayed one "O" or any class introduced after storage (no distillation). Its seeds are not complete; until they are, the evidence is the label-side repairs we ran earlier, all legacy runs (†, test-split early stopping, which favours them).

Evidence so far: partly. Restricting the replay loss to the classes alive at storage gives 39.4 AA (seed 42); restricting the current loss as well, so that a current "O" token is trained only against "O" and the new classes, gives 60.5 on average over three seeds; adding model pseudo-labels on both sides, as the pseudo-labelling methods do, gives 44.1 (one seed). No repair exceeds 69.3 AA on any seed, below every seed of CoLaR-SBR. In the relabelled-memory run old-session recall stays at 77.2%, against 88.4 to 92.6% for CoLaR-SBR (Tab. 2); the current documents' pull is the remaining

suspect, and the MiB-style control measures how much of it a corrected loss removes.

4.4. H3: is it the shared normalisation?

Prediction. If the conflict lives in the shared normalisation, one background logit trained by documents that disagree about what is background, then one background logit per session, normalised with that session's tags only, should close most of the gap with no true labels. This is SS-IL's diagnosis of a shared softmax [1] with a background added, and ICE-O's readout [26] without its frozen encoder. It needs the memory: separating the labels does not undo drift, so without a memory the same readout should fail, and fail the way drift does.

Result: supported, on every seed. With the memory the session-blocked readout (CoLaR-SBR, Sec. 5.1) reaches 89.1±2.0 full-label F1 and 88.4±2.6 AA, 0.6 to 5.3 full-label points below joint training per seed and above the relabelled memory on every seed both have, under both metrics. The owning block claims 95.4 to 98.3% of the old entity tokens, and most remaining errors (56.6 to 71.6% of the wrong tokens) are confusions inside a session. Without the memory the readout scores 11.4±2.3 (21.6±2.2 AA), with the other signature: not one old entity token is predicted "O", 83.8 to 93.5% of the wrong ones receive another session's class, and each old block claims 17.0 to 24.3% of the tokens of its own split that are not its entities, against 0.6 to 1.7% with the memory. Old blocks on a drifted trunk claim far more than their own entities; that is drift, and the memory removes it.

The conflict on one receipt. Take a receipt stored at the end of session 1, when only menu fields exist: its item prices are labelled `menu.price` and its total price "O". In session 5 a current receipt labels its total price `total.total_price` and its item prices "O", since menu fields are not annotated in that session. In one batch the two supervise the one background logit of a shared softmax on look-alike price strings, the stored receipt pulling totals towards "O" and the current one pulling item prices towards it. The refresh removes the first pull for the documents stored before the last boundary; the current documents keep the second. Under SBR the stored receipt trains only block 1 and the current one only block 5, so neither pulls the other's background.

What goes against it. The readout does nothing alone, and it is not free: on DIL, where there is nothing to separate, CoLaR-SBR is below CoLaR (AA 86.2 against 87.8†).

What the measurements do not show. The shared-softmax runs saved per-class reports, not per-token outputs, so their errors are not split by type. Block measurements use session-masked splits whose "O" tokens include other sessions' entities, so we count only claims by a split's own block, wrong whatever the token is.

5. From diagnosis to design

The principle. The rungs point at one fact about annotation under label growth, familiar from training on partially annotated data [11]. A token marked "O" in session s means "none of the classes annotated in session s": it is evidence against those classes and silent about every other. A shared softmax turns it into evidence against all classes at once (H3), and stale stored labels are the same error made by the memory (H1). The principle: *supervise each class only with negatives from documents that annotated it, and normalise together only classes that were annotated together.* Three consequences follow: each session needs its own background; the memory must keep whole documents with their own labels and the session that wrote them; and each session's background logit, trained on that session's true negatives, is also a detector of its entities, so it can route a token without task identity.

5.1. SBR: one background per session

SBR is our instantiation of a per-session background readout [19, 26]. Every session that brings new tags opens a block b holding those tags $\mathcal{T}_b$. A shared softmax normalises over $\{\mathrm{O}\} \cup \bigcup_b \mathcal{T}_b$. SBR gives every session its own background logit $o_b(x) = w_b^\top h(x) + c_b$ on the token feature $h(x)$ and normalises each block separately:

$$p_b(\cdot \mid x) = \mathrm{softmax}\left([\, o_b(x),\, z_{\mathcal{T}_b}(x)\,]\right), \tag{1}$$

where $z(x)$ are the tag logits of the usual linear head and $z_{\mathcal{T}_b}(x)$ its entries for the tags in $\mathcal{T}_b$. Within block b a token labelled "O" in session-b data is truly background, so the block's supervision is never stale. For a token x with label y in a document of session s,

$$\mathcal{L}(x, y, s) = \sum_b m_{sb}\, \mathrm{CE}\big(p_b(\cdot \mid x),\, y_b\big), \qquad y_b = \begin{cases} y & y \in \mathcal{T}_b \\ \mathrm{O}_b & \text{otherwise,} \end{cases} \tag{2}$$

with one rule for the mask: $m_{sb}{=}1$ when session s annotates every tag of $\mathcal{T}_b$; $m_{sb}{=}1$ with $y_b{=}\mathrm{O}_b$ when b belongs to a different annotation schema (the document is annotated under its own schema, so every token is a negative for b; a training-time convention that, between related schemas such as SROIE and CORD, can call a real entity background); and $m_{sb}{=}0$ otherwise, because the document's "O" labels say nothing about block b. Stored documents follow the rule of the session that stored them. A session that adds no

tags creates no block, so on a fixed label space SBR has a single block. At inference each block reports its claim $e_b = 1 - p_b(\mathrm{O}_b \mid x)$; the block with the largest e_b labels the token with its best tag if $e_b > 0.5$, and the token is "O" otherwise, as in ICE-O's "Other unless some session scores above it". Blocks trained at different times need not produce claims on one scale, which is why Tab. 2 measures the routing directly. App. C gives the pseudo-code, the supervision mode of every (session, block) pair and the memory accounting.

5.2. The memory

SBR's supervision needs stored documents that keep each token's label at its position and the session that wrote it. The memory we use, CoLaR (Compressed Latent Replay), keeps whole documents and compresses each one on its own. The pooled layer-k feature space is close to full rank, but one document's tokens share a layout and a vocabulary, so its activation matrix $H_j \in \mathbb{R}^{N_j \times d}$ is low-rank; CoLaR stores $H_j \approx U_r \Sigma_r V_r^\top$ with the document's boxes, mask, labels and session, rebuilds $\hat{H}_j$ at replay and injects it at layer k. On DIL it retains $87.8{\pm}0.5$ AA in 60 MB (1 MB $= 10^6$ bytes), against $87.9{\pm}1.4$ for ER with 200 raw documents, and it holds no raw pixels or token ids (no privacy claim: activations can leak content). Summaries of the same store injected below the self-attention layers (pooled spectral summaries, per-class Gaussians, centroids) fall to the naive level, as per-class Gaussian latent replay does on images; a class statistic used at the readout works on images [56], while the readout-level Gaussian realignment we ran on documents (LCA's, after merging) does not repair drift (App. F, which holds the ladder, the compression trade-off and the image comparison).

5.3. SBR-Recall: the test of H4

Prediction (H4). With the conflict removed, the memory is needed only against drift, and the task-identity view of class-incremental learning [20] says when it can go. If each block is read under the trunk state that learned it, nothing drifts for that block and the memory could go, provided the model can tell which state a token belongs to. On one schema under label growth each block's background logit supplies that key. Across schemas the all-background terms of Eq. (2) train every old block towards background on other-schema documents with nothing to pull it back, so isolation should fail without stored documents; on a fixed label space there is one block and nothing to route by.

The intervention. SBR-Recall stores, for each session $b > 1$, a low-rank delta Δ_b [16] of the layers above k (rank 64 on the attention projections and feed-forward layers), freezes these layers after the first session (the base θ_0; $\Delta_1{=}0$), and computes block b under the trunk state that learned it:

$$p_b(\cdot \mid x) = \mathrm{softmax}\left([\, o_b,\, z_{\mathcal{T}_b}\,]\right)\left(x;\, \theta_0 + \Delta_b\right). \qquad (3)$$

Every term of Eq. (2) is computed under the delta its block is read under, at training and test time. Without a memory on a single-schema stream no term of a later session involves an old block, so each old block's e_b is unchanged up to weight decay, and only competition from new blocks can move its tokens (App. C). A session that adds no block reuses the delta of the first block it trains fully.

Table 4. **Comparison on LayoutLMv3** (full-label F1 and session-restricted AA on CIL-CORD; AA on Mixed and DIL; mean ± s.d. over three seeds, per-seed values when fewer; "–": not run). †legacy protocol; $^{n=k}$: k clean seeds so far; ^{c}one fp16 LwF seed that collapsed to all-background is excluded; ^{b}with the 50-document memory. Settings per cell: App. G.

Method	CIL-CORD full-label F1	CIL-CORD AA	Mixed AA	DIL AA
Naive	13.0±0.2	18.3±0.5	44.4/43.7†	41.3±1.5†
ER	–	24.3±2.4†	75.6/73.3†	87.9±1.4†
LwF	–	18.4/18.5†,c	47.7†,c	43.7±1.1†
DER++	–	20.4±1.4†	75.1/74.0†	88.0±0.9†
CODA-Prompt	–	14.0/15.9†	12.1/10.6†	32.5±1.2†
CL-LoRA (2025)	–	18.7/18.6†	45.7/43.9†	41.1±0.3†
CPFD (2023)	–	2.7/0.0†	0.3/15.1†	6.1±7.3†
IS3 (2024)	–	5.6/7.7†	0.0/40.6†	38.3†
LCA (2026)	–	13.7/7.0†	38.0/36.4†	43.9±2.5†
CoLaR	30.7±1.5	33.5±1.0	75.6/74.6†	87.8±0.5†
CoLaR-SBR	89.1±2.0	88.4±2.6	**89.8**/**89.6**†	86.2±1.3†
+ LexSlot head slots	–	89.7±0.5†	89.6±1.0†	87.0±0.4†
SBR-Recall	**92.6±1.0**	**91.9±0.6**	85.9†,b	–
Joint training (reference)	92.0±0.4	91.3±0.8	86.9/88.5†	88.7±1.1†

Result: supported, with its predicted failure. On CIL-CORD SBR-Recall reaches $92.6{\pm}1.0$ full-label F1 ($91.9{\pm}0.6$ AA) with no stored document, within 2.1 points of joint training on every seed under both metrics; its owning block claims 97.0 to 98.3% of the old entity tokens. On Mixed without a memory it scores 44.7 AA† (seed 42): the blocks of the other schemas fall silent until their domain returns, as predicted, and with the 50-document memory restored it scores 85.9 (App. D). On a fixed label space SBR-Recall has a single block; that it cannot replace the memory there follows from the design and was not run. We ran no shared-softmax counterpart that combines or merges the deltas (as EASE and InfLoRA do); our CL-LoRA port shares one adapter across sessions.

6. Evidence beyond one stream

The comparison set. Table 4 reports the fixed comparison set. On CIL-CORD every continual method with a shared softmax stays at or below CoLaR in AA (33.5), and the two blocked readouts close most of the gap: CoLaR-SBR

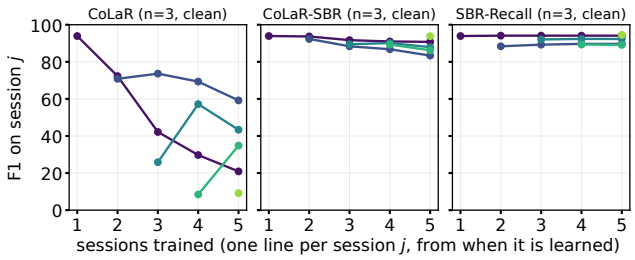


Figure 1. **Retention per session on CIL-CORD** (session-restricted F1 on session j after each later session, mean over the seeds; panel titles give their number n and the protocol, † = legacy). With a shared softmax the curves fall steeply and late sessions are learned poorly; with the blocked readout most sessions hold; under SBR-Recall no session falls more than 0.6 points below its diagonal on any seed.

89.1±2.0 and SBR-Recall 92.6±1.0 full-label F1 against 92.0±0.4. Four baselines end below naive training on every seed because they learn later sessions poorly: CPFD learns no session after the first (at most 0.0 F1 on a later diagonal), and IS3, CODA-Prompt and LCA each learn some later session on every seed to at most 32.3 F1, where naive training learns every session to at least 78.6; like naive training they keep little of the earlier sessions (App. G). These rows are our ports onto one growing token head, not checked against their published NER numbers. On Mixed, where both causes act, CoLaR-SBR reaches 89.7 AA against 87.7 for joint training and 74.5 for ER. On DIL, where only drift is present, CoLaR, CoLaR-SBR and ER all come close to joint training on average (AA 87.8, 86.2, 87.9 against 88.7). App. D lists the per-seed values behind every cell.

Where the remaining gap lies. With a shared softmax the curves fall steeply and irregularly, some sessions partly recovering at a later boundary, and the later sessions are learned poorly even on their own diagonal (Fig. 1). With the blocked readout every session is learned and most keep nearly all of their F1, with one early session losing the most. Under SBR-Recall the curves are flat: no session loses more than 0.6 points on any seed, and the larger moves, up to 2.1, are gains (App. D).

Ablations. With every encoder layer but the last frozen after the first session, and no memory, the blocked readout scores 14.8 AA† (one seed; sessions after the second are barely learned): the blocks need both a plastic trunk and a memory that keeps it stable (App. E). Tuning the claim threshold and a per-block bias on the test set itself changes AA by at most 0.6 (seed 42), so what SBR still loses to joint training is retention of early sessions, the part SBR-Recall removes. For SBR-Recall without a memory, rank 8 on the attention projections alone gives 82.5 AA (seed 42) with flat retention but two sessions stuck at 65 and 68 F1; rank 64 with the feed-forward layers learns them (App. D). Stacking LexSlot head slots on CoLaR-SBR moves the mean by less than two points on every stream (App. D).

Cost. CoLaR-SBR stores 50 compressed documents per session (100.7 MB on CIL-CORD) and one background logit per session. SBR-Recall stores the deltas of the four later sessions (57 MB in half precision, 113 MB in fp32 as trained) and pays one upper-layer pass per session at inference (about 3.7× the encoder compute at five sessions; App. C, H).

7. Limitations and conclusion

Limitations. The study covers three English datasets with test sets of 50 to 347 documents and four encoder-only backbones; the ladder, SBR and SBR-Recall are evaluated on LayoutLMv3 and on one class-incremental partition of CORD. The freezing experiment, the memory-content ladder, the relabelled memory (one seed so far) and SBR-Recall on Mixed are single- or few-seed diagnostics. Comparators from this literature that we did not run: a faithful ICE-O (frozen encoder, one head per session, constant Other logit), SBR with a constant background, a separated softmax with one shared background, per-class sigmoid heads (SSUL, SpanKL), MiB's distillation, PLOP and BPF; the MiB-style loss of Sec. 4.3 is the background-shift correction we run on the same memory. Joint training is a reference, not a bound. SBR-Recall grows by one delta and one pass per session and freezes its base after session 1, so it may depend on the session order. SBR's rule reads each session's schema at training time. CoLaR's memory grows with the sessions, and storing no raw pixels or token ids is weaker than privacy.

Conclusion. Replay retains documents under domain shift and fails under label growth because two things go wrong, and a memory repairs only one. Drift is answered by a whole-document memory. The background conflict is not: stale stored labels are part of it, and relabelling the older stored documents with their true labels leaves a shared softmax below joint training and below CoLaR-SBR (one seed). It is answered at the readout, by letting an "O" label count only against the classes its annotator could see, and where label growth supplies the routing key that readout lets per-session deltas replace the memory. Continual document methods should be tested on a growing label space, with a mask-free metric, against a memory whose stale labels are refreshed.

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